

# 4D Decomposition

Overlapping Rules → Disjoint Cells → TRF

## Input LRF (overlapping)

r1: IP ANY → ALLOW

r2: TCP 10.0.0.x / 80-1023 → DENY (correlation conflict)

↓ *sweep-line 3D*

## After 4D Decomposition (disjoint cells)

**TCP · src≠10.0.0.x**

port 0-1023

→ **ALLOW**

**TCP · 10.0.0.x**

port 0-79

→ **ALLOW**

**TCP · 10.0.0.x**

port 80-1023

→ **DENY**

**UDP / ICMP · ANY**

(inherited)

→ **ALLOW**

↓ *directional merge*

## TRF Output

[protocol] = TCP

[dst\_port]

0-79  
**ALLOW**

80-1023  
**DENY**

## Conversion algorithm (4 stages)

① Protocol  
expansion

② Sweep-  
Line 3D

③ Directional  
merge

④ Projection  
normalization